



UNIVERSITY OF PERADENIYA
FACULTY OF SCIENCE
SEMESTER II - 2019/2020
MT 105 - REAL ANALYSIS I
END SEMESTER EXAMINATION

Time allowed: **THREE** hours

Instructions.

- There are **FOUR** questions on **THREE** pages. Answer **ALL** the questions.
 - Clearly justify each and every step of your solutions with reasoning to receive the assigned number of points. Whenever needed, you can “quote” the results as learned in class without proving.
 - The notations and terms are as used in the class, especially $\mathbb{N} = \{1, 2, 3, \dots\}$ denote the set of natural numbers.
 - Calculators are not allowed.
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1. State the **Completeness Axiom** and the **Archimedean Principle**.

- (a) By clearly justifying your reasoning, find the infimum and supremum of the set

$$E = \left\{ 2 - \frac{(-1)^n}{n^2} : n \in \mathbb{N} \right\}.$$

- (b) Suppose that $A \subseteq B$ are non-empty subsets of $\mathbb{R}$.

- If B has a supremum, then show that $\sup A \leq \sup B$.
- If B has an infimum, then show that $\inf A \geq \inf B$.

- (c) Prove the statement for the **Density of rational numbers** : If $a, b \in \mathbb{R}$ satisfy $a < b$, then there is a $q \in \mathbb{Q}$ such that $a < q < b$.

Using the previous result, state and deduce an analogous result for **irrational numbers**.

- (d) If $x > 0$, show that there exists $n \in \mathbb{N}$ such that $\frac{1}{2^n} < x$.

(100 points)

[Turn over

2. (a) Use the $\epsilon - N$ definition to show that $\lim_{n \rightarrow \infty} \frac{3n+1}{7n-4} = \frac{3}{7}$, by providing a suitable $N \in \mathbb{N}$ value.

(b) Let $\{x_n\}$ and $\{y_n\}$ be sequences of real numbers. Assume that x_n is bounded for all $n \in \mathbb{N}$ and $y_n \rightarrow 0$ as $n \rightarrow \infty$. Show that the product sequence $\{x_n y_n\}$ converges to zero.

(c) State and prove the **Monotone Convergence Theorem**.

Consider the sequence recursively defined by $x_1 = 2$ and

3.
$$x_{n+1} = 2 - \frac{1}{x_n} \text{ for } n \in \mathbb{N} \setminus \{1\}.$$

i. Show that $x_n \geq 1$ for all $n \in \mathbb{N}$.

ii. Show that $\{x_n\}$ is a decreasing sequence.

iii. Does $\lim_{n \rightarrow \infty} x_n$ exist? If so, find the limit. If not, explain the reason.

(d) Find $\lim_{n \rightarrow \infty} \frac{1}{4^n} \cos(n^3 - n^2 + n - 2022)$.

(100 points)

3. (a) Use the $\epsilon - \delta$ definition to show that $\lim_{x \rightarrow 0} x^3 \sin(e^{x^2}) = 0$.

(b) The *Dirichlet function* is defined on $\mathbb{R}$ by

$$f(x) := \begin{cases} 1 & \text{if } x \in \mathbb{Q}, \\ 0 & \text{if } x \notin \mathbb{Q}. \end{cases}$$

Prove that every point $x \in \mathbb{R}$ is a point of discontinuity of f .

(c) State and prove the **Extreme Value Theorem (EVT)**.

i. With reference to the EVT, provide an example and clearly justify why the *closed* assumption on the interval cannot be dropped.

ii. With reference to the EVT, provide an example and clearly justify why the *bounded* assumption on the interval cannot be dropped.

(d) State the **Intermediate Value Theorem (IVT)**.

If $f : [a, b] \rightarrow [a, b]$ is continuous, then show that f has a fixed point; that is, there is a point $c \in [a, b]$ such that $f(c) = c$.

(100 points)

4. (a) Consider the function

$$f(x) = \begin{cases} \sin(1/x) & \text{if } x \neq 0, \\ 0 & \text{if } x = 0. \end{cases}$$

Show that f is not differentiable at 0. (Hint: Differentiable implies continuous.)

- (b) Consider the function

$$g(x) = \begin{cases} x^2 \sin(1/x) & \text{if } x \neq 0, \\ 0 & \text{if } x = 0. \end{cases}$$

Show that g is differentiable at 0 and $g'(0) = 0$.

- (c) State the **Rolle's theorem**.

- i. With reference to the Rolle's theorem, provide an example and clearly justify why the *continuous* assumption of the function cannot be dropped.
- ii. With reference to the Rolle's theorem, provide an example and clearly justify why the *differentiable* assumption of the function cannot be dropped.

- (d) State and prove the **Mean Value Theorem** (MVT).

Suppose that f is twice differentiable on (a, b) and that there are points $x_1 < x_2 < x_3$ in (a, b) such that $f(x_1) > f(x_2)$ and $f(x_3) > f(x_2)$. Prove that there is a point $c \in (a, b)$ such that $f''(c) > 0$.

(100 points)